DATABASE MANAGEMENT LAB Report

Submitted in partial fulfilment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

to

K.R Mangalam University

Submitted by:

RADHIKA

(2401010094)

Submitted to

Ms. Mansi Kajal

(Assistant Professor)

Department of Computer Science and Engineering School of Engineering and Technology
K.R Mangalam University, Gurugram- 122001, India SESSION – 2025-2026

LAB ASSGINMENT 1

Objective:

- To design ER diagrams and convert entities and relationships into relational tables using proper schema notation.

Instructions:

1. Open your preferred Database Management System (DBMS) tool.
2. Create a new database named DBMS_YourName.
3. Draw ER diagrams for the given scenarios.
4. Convert entities and relationships into relational schemas.
5. Create tables using the derived schemas.

Relational Schema

PART A – COLLEGE DATABASE

STUDENT (USN, SName, Address, Phone, Gender)
SEMSEC (SSID, Sem, Sec)
CLASS (USN, SSID)
SUBJECT (Subcode, Title, Sem, Credits)
IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

```
CREATE DATABASE DBMSLAB_1_Radhika;
USE DBMSLAB_1_Radhika;
#PART A COLLEGE DATABASE
CREATE TABLE STUDENT (
    USN VARCHAR(20) PRIMARY KEY,
    SName VARCHAR(50),
    Address VARCHAR(100),
    Phone VARCHAR(15),
    Gender VARCHAR(10)
);
CREATE TABLE SEMSEC (
    SSID INT PRIMARY KEY,
    Sem INT,
    Sec VARCHAR(5)
);
CREATE TABLE CLASS (
    USN VARCHAR(20),
    SSID INT,
    PRIMARY KEY (USN, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
CREATE TABLE SUBJECT (
    Subcode VARCHAR(10) PRIMARY KEY,
```

```
    Address VARCHAR(100),
    Phone VARCHAR(15),
    Gender VARCHAR(10)
);
CREATE TABLE SEMSEC (
    SSID INT PRIMARY KEY,
    Sem INT,
    Sec VARCHAR(5)
);
CREATE TABLE CLASS (
    USN VARCHAR(20),
    SSID INT,
    PRIMARY KEY (USN, SSID),
    FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
CREATE TABLE SUBJECT (
    Subcode VARCHAR(10) PRIMARY KEY,
    Title VARCHAR(50),
    Sem INT,
    Credits INT
);
```

PART B – COMPANY DATABASE

EMPLOYEE (SSN, Name, Address, Sex, Salary, Superssn, Dno)
DEPARTMENT (Dno, Dname, Mgrssn, Mgrstartdate)
DLOCATION (Dno, Dloc)

PROJECT (PNo, PName, PLocation, DNo)
WORKS_ON (SSN, PNo, Hours)

```sql
);
#PART B COMPANY DATABASE
CREATE TABLE EMPLOYEE (
    SSN INT PRIMARY KEY,
    Name VARCHAR(50),
    Address VARCHAR(100),
    Sex VARCHAR(10),
    Salary INT NOT NULL,
    Superssn INT,
    Dno INT
);
CREATE TABLE DEPARTMENT (
    Dno INT PRIMARY KEY,
    Dname VARCHAR(50),
    Mgrssn INT,
    Mgrstartdate DATE
);
CREATE TABLE DLOCATION (
    Dno INT,
    Dloc VARCHAR(50),
    PRIMARY KEY (Dno, Dloc),
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);
```

```sql
);
CREATE TABLE DLOCATION (
    Dno INT,
    Dloc VARCHAR(50),
    PRIMARY KEY (Dno, Dloc),
    FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);
CREATE TABLE PROJECT (
    PNo INT PRIMARY KEY,
    PName VARCHAR(50),
    PLocation VARCHAR(50),
    DNo INT,
    FOREIGN KEY (DNo) REFERENCES DEPARTMENT(Dno)
);
CREATE TABLE WORKS_ON (
    SSN INT,
    PNo INT,
    Hours INT,
    PRIMARY KEY (SSN, PNo),
    FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN),
    FOREIGN KEY (PNo) REFERENCES PROJECT(PNo)
);
```

PART C – CUSTOMER SALE SCHEMA

Database Schema for a customer-sale scenario

Customer(Cust id : integer, cust_name: string)

Item(item_id: integer, item_name:string, price: integer)
Sale(bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)

```sql
#PART C – CUSTOMER SALE SCHEMA
CREATE TABLE CUSTOMER (
    Cust_id INT PRIMARY KEY,
    Cust_name VARCHAR(50)
);
CREATE TABLE ITEM (
    Item_id INT PRIMARY KEY,
    Item_name VARCHAR(50),
    Price INT NOT NULL
);
CREATE TABLE SALE (
    Bill_no INT PRIMARY KEY,
    Bill_date DATE,
    Cust_id INT,
    Item_id INT,
    Qty_sold INT,
    FOREIGN KEY (Cust_id) REFERENCES CUSTOMER(Cust_id),
    FOREIGN KEY (Item_id) REFERENCES ITEM(Item_id)
);
```

DDL Questions (Data Definition Language):

2-Write SQL statements to create all the above relations with appropriate PRIMARY KEY and FOREIGN KEY constraints.

```sql
    Bill_date DATE,
    Cust_id INT,
    Item_id INT,
    Qty_sold INT,
    FOREIGN KEY (Cust_id) REFERENCES CUSTOMER(Cust_id),
    FOREIGN KEY (Item_id) REFERENCES ITEM(Item_id)
);
#DDL Questions (Data Definition Language):
#QUES_2
ALTER TABLE STUDENT
ADD Email VARCHAR(50);
SELECT * FROM STUDENT;
```

USN	SName	Address	Phone	Gender	Email
NULL	NULL	NULL	NULL	NULL	NULL

3-Apply NOT NULL constraints to Salary and Price attributes.

ANS-Alredy applied in above table.

Salary INT NOT NULL

Price INT NOT NULL

4-Rename attribute SName to StudentName in STUDENT relation.

```
FOREIGN KEY (Cust_id) REFERENCES CUSTOMER(Cust_id),
FOREIGN KEY (Item_id) REFERENCES ITEM(Item_id)
);
#DDL Questions (Data Definition Language):
#QUES_2
ALTER TABLE STUDENT
ADD Email VARCHAR(50);
SELECT * FROM STUDENT;
#QUES_4
ALTER TABLE STUDENT
RENAME COLUMN SName TO StudentName;
SELECT * FROM STUDENT;
```

USN	StudentName	Address	Phone	Gender	Email
NULL	NULL	NULL	NULL	NULL	NULL

DML Questions (Data Manipulation Language):

1-Insert at least 10 tuples into CUSTOMER, ITEM, and SALE relations.

```
INSERT INTO CUSTOMER VALUES
(1,'Aman'),
(2,'Riya'),
(3,'Rahul'),
(4,'Neha'),
(5,'Simran'),
(6,'Arjun'),
(7,'Karan'),
(8,'Meera'),
(9,'Rohit'),
(10,'Anjali');
SELECT * FROM CUSTOMER;
```

Cust_id	Cust_name
1	Aman
2	Riya
3	Rahul
4	Neha
5	Simran
6	Arjun
7	Karan
8	Meera
9	Rohit
10	Anjali

```
INSERT INTO ITEM VALUES
(101,'Laptop',50000),
(102,'Mouse',500),
(103,'Keyboard',1500),
(104,'Headphones',2000),
(105,'Monitor',12000),
(106,'Tablet',15000),
(107,'Charger',800),
(108,'Camera',30000),
(109,'Printer',7000),
(110,'Speaker',2500);
SELECT * FROM CUSTOMER;
```

Cust_id	Cust_name
1	Aman
2	Riya
3	Rahul
4	Neha
5	Simran
6	Arjun
7	Karan
8	Meera
9	Rohit
10	Anjali
NULL	NULL

```
INSERT INTO SALE VALUES
(1001,CURDATE(),1,101,1),
(1002,CURDATE(),2,102,2),
(1003,CURDATE(),3,103,1),
(1004,CURDATE(),4,104,3),
(1005,CURDATE(),5,105,1),
(1006,CURDATE(),5,110,2),
(1007,CURDATE(),6,108,1),
(1008,CURDATE(),7,109,1),
(1009,CURDATE(),8,106,2),
(1010,CURDATE(),9,107,4);
SELECT * FROM SALE;
```

Bill_no	Bill_date	Cust_id	Item_id	Qty_sold
1001	2026-03-01	1	101	1
1002	2026-03-01	2	102	2
1003	2026-03-01	3	103	1
1004	2026-03-01	4	104	3
1005	2026-03-01	5	105	1
1006	2026-03-01	5	110	2
1007	2026-03-01	6	108	1
1008	2026-03-01	7	109	1
1009	2026-03-01	8	106	2
1010	2026-03-01	9	107	4

2-List all bills of the current date with customer names and item names.

```
SELECT S.Bill_no, S.Bill_date, C.Cust_name, I.Item_name
FROM SALE S
JOIN CUSTOMER C ON S.Cust_id = C.Cust_id
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE S.Bill_date = CURDATE();
```

Bill_no	Bill_date	Cust_name	Item_name
1001	2026-03-01	Aman	Laptop
1002	2026-03-01	Riya	Mouse
1003	2026-03-01	Rahul	Keyboard
1004	2026-03-01	Neha	Headphones
1005	2026-03-01	Simran	Monitor
1006	2026-03-01	Simran	Speaker
1007	2026-03-01	Arjun	Camera
1008	2026-03-01	Karan	Printer
1009	2026-03-01	Meera	Tablet
1010	2026-03-01	Rohit	Charger

3-Display total bill amount using quantity and price.

```
SELECT S.Bill_no, (I.Price * S.Qty_sold) AS Total_Amount
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Bill_no	Total_Amount
1001	50000
1002	1000
1003	1500
1004	6000
1005	12000
1006	5000
1007	30000
1008	7000
1009	30000
1010	3200

4-List customers who purchased products priced above 200.

```
SELECT DISTINCT C.Cust_name
FROM CUSTOMER C
JOIN SALE S ON C.Cust_id = S.Cust_id
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE I.Price > 200;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Cust_name
Aman
Riya
Rahul
Neha
Simran
Meera
Rohit
Arjun
Karan

5-Count the number of products purchased by each customer.

```
SELECT C.Cust_name, COUNT(S.Item_id) AS Total_Products
FROM CUSTOMER C
JOIN SALE S ON C.Cust_id = S.Cust_id
GROUP BY C.Cust_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Cust_name	Total_Products
Aman	1
Riya	1
Rahul	1
Neha	1
Simran	2
Arjun	1
Karan	1
Meera	1
Rohit	1

6-List products purchased by customer having Cust_id = 5.

```
SELECT I.Item_name
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id
WHERE S.Cust_id = 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Item_name
Monitor
Speaker

7-Create a VIEW showing bill_no, bill_date, cust_id, item_id, price, qty,sold, amount.

```sql
CREATE VIEW Bill_View AS
SELECT S.Bill_no, S.Bill_date, S.Cust_id,
       S.Item_id, I.Price, S.Qty_sold,
       (I.Price * S.Qty_sold) AS Amount
FROM SALE S
JOIN ITEM I ON S.Item_id = I.Item_id;
```

Consider following databases and draw ER diagram and convert entities and relationships to relation table for a given scenario:

COLLEGE DATABASE:

STUDENT (USN, SName, Address, Phone, Gender)

SEMSEC (SSID, Sem, Sec)

CLASS (USN, SSID)

SUBJECT (Subcode, Title, Sem, Credits)

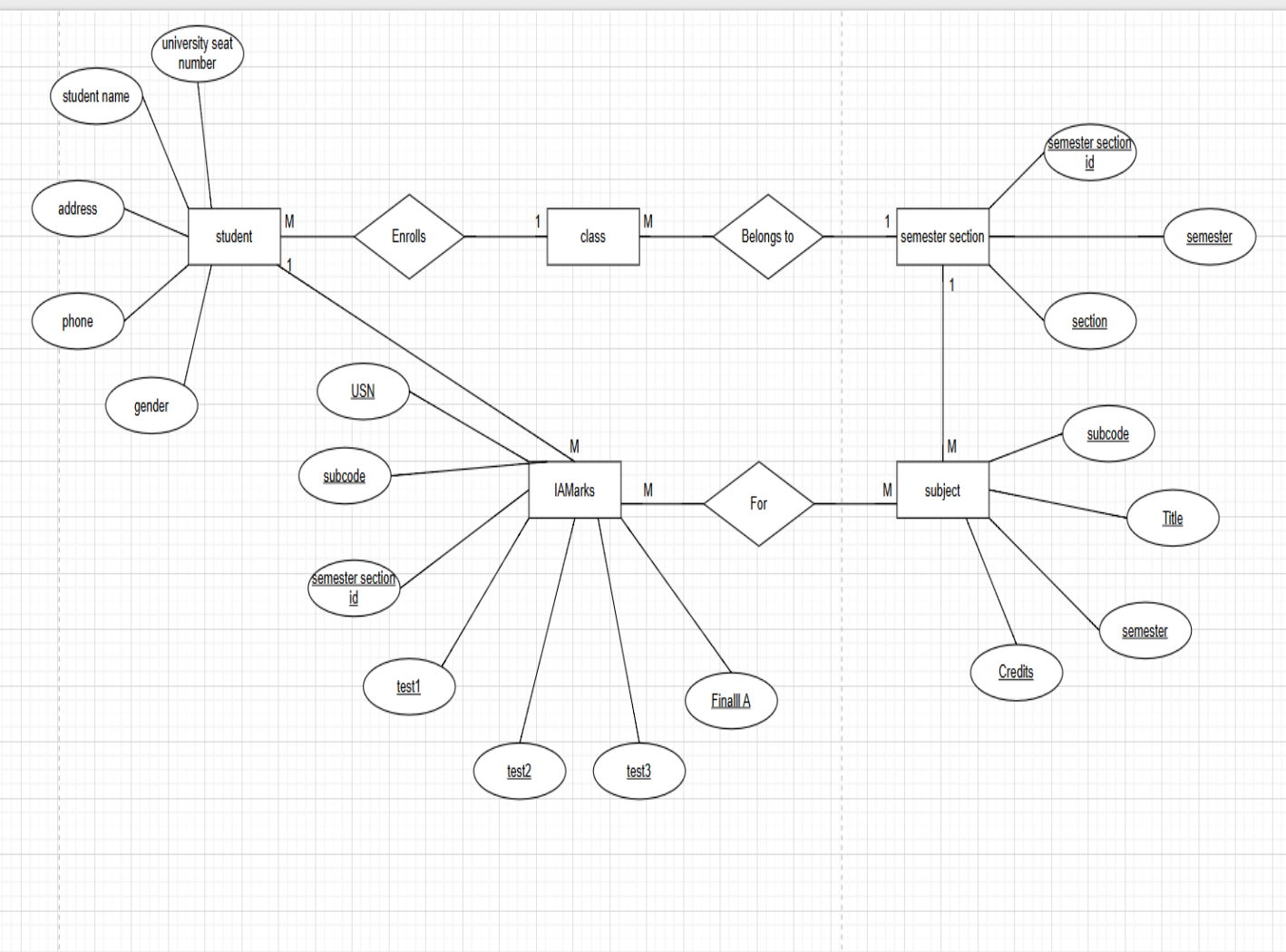
IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

Solution:

1. The strong entities STUDENT, SEMSEC, and SUBJECT are identified and represented using rectangles.
2. The associative entity IAMARKS is identified to resolve the M : N relationship between STUDENT and SUBJECT and is also represented using a rectangle.
3. The attributes are represented using ellipses:

- STUDENT: *USN* (Primary Key), SName, Address, Phone No, Gender Phone No is shown using a double ellipse as it is a multivalued attribute.
- SEMSEC: *SSID* (Primary Key), Sem, Sec
- SUBJECT: *Subcode* (Primary Key), Title, Sem, Credits
- IAMARKS: USN, Subcode, SSID, Test1, Test2, Test3, FinalIA

4. Primary keys are represented by underlining the attribute names.

5. The composite primary key of IAMARKS is (USN, Subcode, SSID).
6. CLASS is drawn as a relationship (diamond) between STUDENT and SEMSEC with M : N cardinality.
7. The relationship between STUDENT and IAMARKS is 1 : M, indicating that one student can have many IA records.
8. The relationship between SUBJECT and IAMARKS is 1 : M, indicating that one subject can have many IA records.
9. The relationship between SEMSEC and IAMARKS is 1 : M, indicating that marks are recorded semester-section wise.
10. All relationships, attributes, primary keys, and cardinalities are clearly marked in the ER diagram.

ER DAIGRAM: